Binary Semantic Segmentation of Dead Trees in Aerial Imagery

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***Abstract*—This report aims to find the most optimal model for segmenting dead trees using aerial images of a single domain. Using a range of methods including ADA-net , U-net and traditional machine learning methods we concluded U-net did best with an intersection of union score of 0.45.**

I. Introduction

In recent years, the health of forests has been on the decline due to factors like deforestation and climate change. Poor forest health causes a wide range of issues. These issues include climate change, increased risk of wildfires, possible loss of biodiversity and soil erosion. Therefore, it is important to actively monitor the health of forests around the world.

The aim of this project is to develop a model that can accurately identify dead trees in aerial images facilitating the monitoring of forest health in a cost-efficient manner. This will be done by exploring a wide range of leading methodologies in computer vision and using 444 aerial images from the US National Agriculture Imagery Program. The results of these methodologies will be compared allowing us to select the most effective method for detecting dead trees in aerial images. This will help facilitate the ability to leverage aerial imagery technology to monitor the health of forests globally.

II. Literature review

Declining forest health remains a significant issue which causes a wide range of problems such as increased risk of wildfires, possible loss of biodiversity and soil erosion. Being able to monitor forest health is critical to restoration efforts. Utilising computer vision techniques offers a potentially cost-effective method of monitoring forest health.

One proposed method for monitoring forest health using aerial multispectral orthoimages is the use of Attention-Guided Domain Adaptation Networks (ADA-net) [Ahishali], which achieved a 4.5% higher DICE score than other best performing domain adaption approaches.

Another leading method that is especially prevalent in medical imaging is U-net. According to a paper published by the University of Technology Sydney, this method is “a staple in medical imaging due to its effective performance and numerous updated versions.”. In this project we will try to utilise U-net for the purpose of segmenting dead trees.

Semantic segmentation models often require some kind of standardisation of image size before processing. Bilinear interpolation is a simplistic but acceptable method of resizing images [Acharya] to suit predetermined backbone dimensions or similar model requirements. The binary segmentation task can also benefit from continuous versions of Intersection-over-Union (IoU) [Rahman] also known as soft Jaccard Loss [Wang], over other loss types such as softmax loss.

Given the limited amount of data in aerial forest imagery of certain biomes [Ahishali], we attempt to improve results by using data augmentation techniques such as rotation and flipping which have been shown to improve results when used on both training and validation sets [Shorten].

We aim to create a simplified version of ADA-net and compare it to U-net, random forest, K-Nearest Neighbours and SVM.

III. Dataset

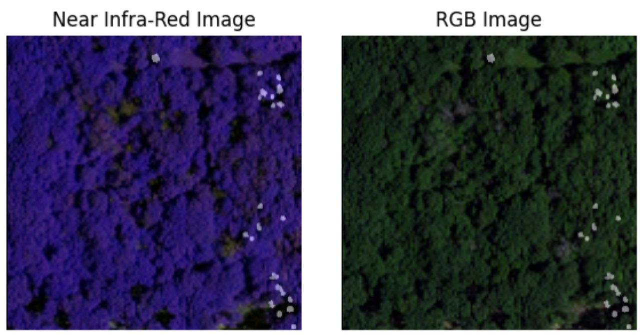

Figure 1: Example visualisation of NRG and RGB images with segmentation mask superimposed

The dataset consists of 444 aerial images of forests from six different states across the U.S., originally captured and made publicly available by the National Agriculture Imagery Program[1] [NAIP] from the United States Department of Agriculture[2] and then curated by Ahishali et

[1] https://naip-usdaonline.hub.arcgis.com/
[2] https://www.usda.gov/

al. [Ahishali]. Each image includes 3 channels of conventional RGB data and a channel of near infra-red transformed into 3 channels. Binary semantic segmentation masks establish the ground truth of dead tree locations within images, based on manual identification by experts. Example images with the segmentation mask superimposed for one location are shown in Figure 1.

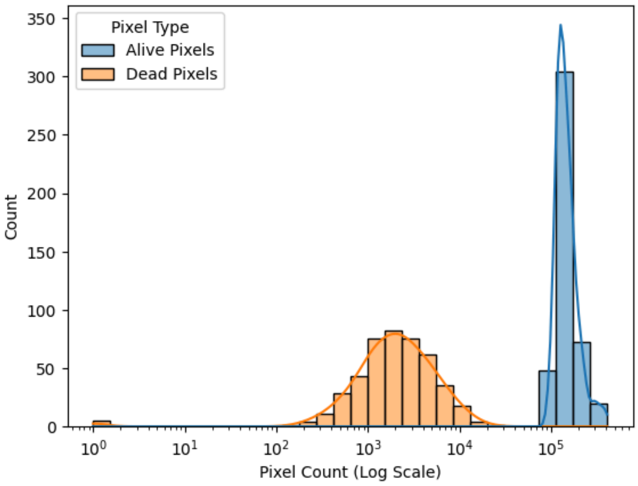

Figure 2: Histogram of alive and dead pixel count per image

The images come in log-normally distributed pixel sizes and with log-normally distributed labelled dead pixel pixel counts, as shown in Figure 2. There is also a significant class imbalance, with the target class representing approximately 1.9% of all pixels in the images. Aspect ratios of the images also varied from 0.90-1.39.

24 images from Washington were determined to be rotated by <10° and zero-padded. These were left untransformed to avoid further distortion.

IV. Methods

A. Traditional Machine Learning Methods

Random Forest (RF) Classifier: The RF algorithm builds an ensemble of decision trees and then aggregates the votes of the trees to create a more stable binary classification (i.e., dead tree, live background). It is a widely renown and ubiquitous classifier in the remote sensing context due to its high accuracy and moderate efficiency i.e, “the Random Forest and Support Vector Machine are usually used for the classification of woody species… [and] can be used to generate detailed maps” [Zagajewski, B]. RF has the distinct advantage that it can perform out-of-bag validation as such making it “relatively robust for outliers and noise, in addition to being computationally lighter than other tree set methods” [Pacheco, A]. In the practical context this results in RF being able to handle many input variables (spectral, indices, textures) without strong assumptions regarding the distribution of the input features. Furthermore, previous works have used RF to map tree health or tree vitality (a context similar to the scope of this project), for example, Shovon *et al.* (2022) used RF to segment alive vs. dead trees from aerial LiDAR and imagery with ~89% accuracy. A key strength in our context, is that as the trees in an ensemble increases there is little risk of "overfitting" and bootstrapping allows the RF algorithm to account for class imbalance. However, careful hyperparameter tuning is still required, with a review noting that “RF is too sensitive to small changes in the training dataset and is occasionally unstable and tends to overfit” [Boateng, E]. In our implementation, we tuned the number of trees and depth through a grid search method. The RF model which produced the best segmentation with an optimized count of trees yielded solid segmentation performance and also fairly rapid predictions on our test images due to the parallel nature of tree-based models.

k-Nearest Neighbors (kNN) Classifier: The kNN method is an instance-based learner, which classifies a pixel based on the majority class of k closest training samples in feature space. This is a non-parametric, simple learning method which makes no assumptions about the statistical distribution of data. This property is valuable for remote sensing problems *“for which there is little or no prior knowledge of data distribution”* [Pacheco, A]. We discovered that a small k (3-5) was acceptable for classification based upon cross-validation. The primary benefit that kNN provided were simplicity and ease of understanding, however, k-NN’s distance metric can struggle with high-dimensional empirical feature vectors and noisy attributes, since every feature contributes equally to the distance calculation. k-NN produced the lowest mean IoU of the three classifiers, reinforcing our observation that its simplified decision boundaries compensate poorly for the spectral-textural patterns characteristic of dead tree canopies.

Support Vector Machine (SVM) Classifier: SVM is a robust maximum-margin classifier with the ability to find a hyperplane that will optimally separate dead tree pixels from those of the background. We used an SVM with a Gaussian RBF kernel, since the pixel classes in question were not linearly separable (the RBF kernel is often recommended *“if the problem might not be linearly separable”* [Boateng, E]). In general, SVMs have been well documented in the remote sensing image analysis field and have been high performers in terms of maximum accuracy. For instance, an SVM with RBF was the best performer in a tree species mapping study, with an F1-score up to ~91% [Zagajewski, B]. Our SVM slightly exceeded RF and gave a very good segmentation IoU on the test set, consistent with the literature stating similar findings. As a result, SVMs can segment new images much faster than kNN. We completed hyperparameter tuning for SVM’s kernel type and C hyperparameter values using GridSearchCV on a validation set to enhance generalization. The best SVM (RBF kernel and medium C) produced smooth segmentation outputs, indicating that the premise that max-margin classifiers are appropriate for this task was valid.

Feature Extraction and Data Preparation: All three classifiers were provided with a comprehensive feature set designed to classify dead trees. The NAIP (National Agriculture Imagery Program) aerial images consisted of four bands (RGB and Near-IR), which we used to derive other indices and features. In particular , we calculated the Normalized Difference Vegetation Index (NDVI) and Green NDVI (GNDVI) for each pixel, which captures vegetation health. Using these spectral indices is known to enhance classification accuracy, – for example, adding NDVI (and NDWI) features enabled a Random Forest to identify tree species with up to 94% accuracy [Zagajewski, B]. GNDVI, which utilizes green instead of red, is particularly effective for detecting vegetation that may be stressed, as using green reflectance circumvents the saturation that occurs when high chlorophyll levels are present “making GNDVI more sensitive to variations in chlorophyll content” [Wu, Q] than standard NDVI. Beyond the vegetation indices, we incorporated the following feature blocks:

<u>Spectral metrics</u>: For each NIR, red, and green band we calculated mean, SD, min, max, and key percentiles.

<u>Texture (LBP)</u>: On grayscale patches we took the mean & SD of Local Binary Pattern values to summarise fine detail.

<u>Edges / gradients</u>:

- Canny edge density captures structural complexity.
- Sobel horizontal + vertical gradients yield magnitude mean & SD.

<u>Intensity moments</u>: Mean, SD, and skewness of grayscale pixels.

<u>Normalisation</u>: All features scaled to zero-mean, unit-variance with *StandardScaler*.

<u>Patches (n x n):</u> Sliding-window sampling; each patch labelled via the mask average → dead vs. live. Local context boosts accuracy over single-pixel classification.

Class Imbalance and Dataset Balancing: A significant challenge was the extreme class imbalance– dead tree pixels were a very vast minority of the pixels in this dataset. If these imbalances weren’t addressed, a classifier could still have high overall accuracy with a high bias just by predicting all pixels as healthy. To address this, we balanced the training data by randomly undersampling the background patches until we had about the same amount of dead-tree and background patches. By keeping every dead-tree example and undersampling the majority class, we gave the classifiers sufficient positive samples to learn the dead-tree signature, without introducing duplicated data.

B. ADA-net

This was the proposed method for the segmentation of dead trees based on a paper published by Cornell University. A simplified version of the model in the paper was implemented with some adjustments. This was due to the data set being from a single domain and not multiple countries.

To prepare the data set for this model. The images were resized to 224 by 224 and normalized between 0 and 1.

The simplified model had three main components. A feature extraction class, an attention module and a segmentation decoder.

The feature extraction class utilises ResNet-50 to try and extract the most relevant features of the image for further processing.

The attention module assigns different weights to different regions of the image to allow the model to highlight key features. The query, key and value of the attention module utilises a 2048 by 2048 convolution layer with kernel size 1. The attention weight is generated by the same function as the one in the paper by Cornell University.

$$\text{Attention}\left(\mathbf{Q},\mathbf{K},\mathbf{V}\right)=\text{SoftMax}\left(\frac{\mathbf{QK}^{T}}{\sqrt{d_q}}\mathbf{V}\right)$$

<u>Figure 3: Histogram of alive and dead pixel count per image</u>

The purpose of the segmentation decoder is to take the key features highlighted by the attention module and translate it to a pixel wise class prediction which segments areas with dead trees that can be compared with the ground truth mask. It is upsized from 7 by 7 back to 224 by 224 using multiple convolution layers.

Using these 3 components, our simplified ADA-net can extract relevant features, highlight key features and then make pixel wise class predictions to compare with the mask.

C. U-Net

For the U-Net method implemented in our project, we used a symmetric encoder-decoder architecture with skip connections, which allows it to effectively capture both context and fine-grained spatial details for semantic segmentation tasks for dead tree detection in aerial imagery.

For encoder path, it follows the original U-Net design with repeated 3×3 convolutions, batch normalization, ReLU activations, and a 2×2 max pooling operation to reduce spatial dimensions. At each downsampling step, the number of feature channels is doubled (64→128→256→512) to capture information. At the network’s deepest point, a bottleneck DoubleConv further expands features to 1024 channels to capture the most abstract representations.

For decoder path, each stage begins with a 2×2 transposed convolution that upsamples feature maps (e.g., 1024→512, 512→256, …, 128→64), each concatenated with its corresponding encoder output via skip connections and refined by another DoubleConv to restore spatial detail.

Finally, a concluding 1×1 convolution shows the 64-channel result to a single-channel mask prediction. No activation is applied at the output layer, as the model directly produces logits and the used loss function, which internally applies the sigmoid function. This setup supports numerical stability and allows for threshold tuning post-training.

We trained the U-Net model using various image standardization strategies, including resizing (128×128, 256×256, 384×384) and patch extraction (32×32, 64×64, 128×128), and evaluated their impact under different data augmentation conditions (rotation and reflection).

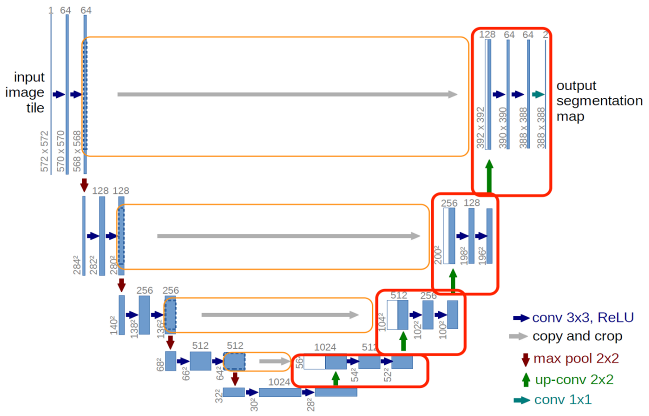

Figure 4: U-Net Encoder–Decoder Architecture with Skip Connections for Semantic Segmentation

D. Data Augmentation and Training for Neural Models

Various strategies can be used to adapt methods that require fixed image sizes, due to restrictions such as fully-connected neural layers. In this study we have experimented with zero-padding, using bilinear interpolation to resize images to fixed sizes and extracting fixed size patches.

Traditional Methods Data Pre-Processing

We loaded the NRG aerial images and their corresponding ground truth masks. Each image was divided into fixed-size square patches using a sliding window approach, with 15 x 15 pixel patch size and with 5 pixels stride. For each patch, we extract a set of features, including statistical, texture, and edge-based information. These features were then used as input to the classifiers. To address class imbalance between healthy and dead tree patches, we applied downsampling to create a balanced dataset containing equal numbers of samples from each class.

Neural Methods Data Pre-Processing

To help address the limited size of the dataset, we experimented with augmented it using rotation and flipping. We used three basic rotations (90°, 180°, 270°) referred to as “Use Rotation” and four transformations involving flipping (horizontal reflection, vertical reflection, transposition and horizontally reflected transposition) referred to as “Use Reflection”. “Use Rotation” increased the dataset size to 4 times its original size, “Use Reflection” to 5 times and both transformations increased the dataset size by 8-fold.

Given the significant right tail in image dimensions, as shown by its log-normal distribution, zero-padding images to standardise their size would result in 656 x 656 pixel images in contrast to the 382 x 372 median pixel dimensions. We employed this approach as a baseline for more appropriate alternatives.

Our adaptation of ADA-Net utilises the ResNet-50 image classifier [He] as a backbone, which has a fixed input requirement of 224 x 224 pixels. Similarly, while our U-Net architecture does not require a fixed image size, efficient batching for training is only possible with similarly sized arrays. To overcome these issues we experimented with bilinear interpolation to resize to 224 x 224 for ADA-Net and 128 x 128, 256 x 256 and 384 x 384 for U-Net. Due to concerns about inflating the IoU, mask predictions were resized back to their original size on the test, to ensure fair comparison. We also experimented with extracting patches to preserve the original image aspect ratio and improve the ratio of model parameters to observations in the dataset. For ADA-Net we extracted 224 x 224 patches, with no stride and with zero-padding. For U-Net, we used patches of size 32 x 32, 64 x 64 and 128 x 128, also with no stride and zero-padding.

All combinations of reflection and rotation were applied to each configuration except zero-padding, resulting in 33 different models.

The dataset was split into 20% testing, and the remaining 80% was further split into 90% training, 10% validation for the neural methods. All splits were stratified by U.S. state in order to avoid further imbalances due to differences in biomes. We used a differentiable version of the Jaccard score, known as soft Jaccard loss [Wang] as our loss function, as it is resilient to class imbalance issues and is directly related to our target performance metric. We utilised a modified early stopping algorithm with a patience of 8 epochs and a minimum decrease in validation loss of 0.01. We also utilised a minimum epochs requirement of 20[3] epochs to help prevent underfitting. The best performing model based on validation loss was saved and used, to avoid overfitting.

Since both models output logits, we further fit the optimum decision threshold by minimising IoU across the training and validation set predictions.

All experiments were run on a Ubuntu Machine with 32Gb of RAM, NVIDIA GeForce RTX 4060 GPU and a 13th Gen Intel® Core™ i9-13905H × 20 CPU and training and testing times were recorded.

V. Experimental Results

A. ADA-net

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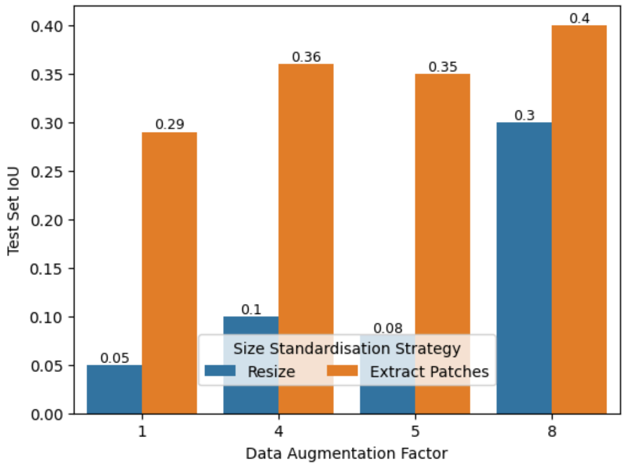

Figure 5: ADA-Net Performance Across Resizing and Patching Strategies with Different Data Augmentation Levels

To evaluate the effectiveness of the ADA-Net method for dead tree detection, we conducted experiments using two size standardisation strategies: bilinear resizing and patch extraction. We varied the data augmentation factor (1, 4, 5, 8) to compare its impact on model performance, training time, and testing efficiency.

For the accuracy part, patch extraction consistently outperformed resizing the images across all augmentation factors. The best Intersection over Union (IoU) of 0.40 was at augmentation factor 8 for patch extraction, compared to 0.30 for resizing at the same level. Moreover, the lowest IoU of 0.05 was achieved by resizing at augmentation factor 1, barely above random pixel selection. However, train IoU for the extract patch methods consistently reached above 0.65, with a peak of 0.69, indicating that it vastly overestimates its performance.

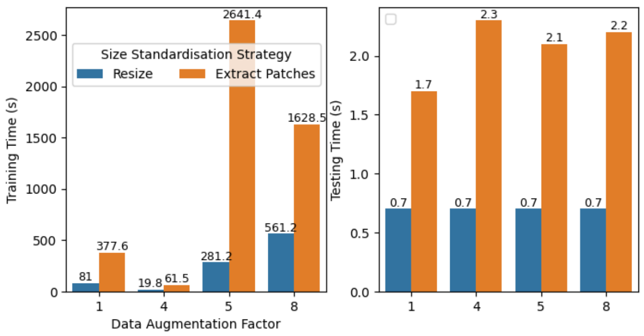

Figure 6: ADA-Net Training and Testing Times

In terms of training and testing time, patch extraction showed significantly higher training time compared to resizing. It has the longest training time reaching 2641.4 seconds at an augmentation factor of 5. On the other hand, resizing was faster to train, the training time was dropping to as low as 19.8 seconds at augmentation factor 4. However, this speed advantage came at the cost of reduced model accuracy.

Testing times were relatively stable across all settings, although resizing remained faster than patch extraction. Models using resizing achieved testing times of approximately 0.7 seconds, while patch extraction got longer, averaging between 2.2 and 2.3 seconds.

B. U-Net

We evaluate the performance of U-net implementation across a set of experiences, it consists of two main image standardization which is bilinear resizing and patch extraction under varying input dimensions and data augmentation levels.

Our baseline performance for U-Net by simply padding to the much larger 656 x 656 resolution, resulted in an IoU of 0.391. Out-of-Memory (OOM) issues made further training zero-padding with augmentation factors greater than one difficult.

The performances are measured by Intersection over Union (IoU) on the test set, and computational efficiency was tracked using training and testing time.

Resizing Strategy:

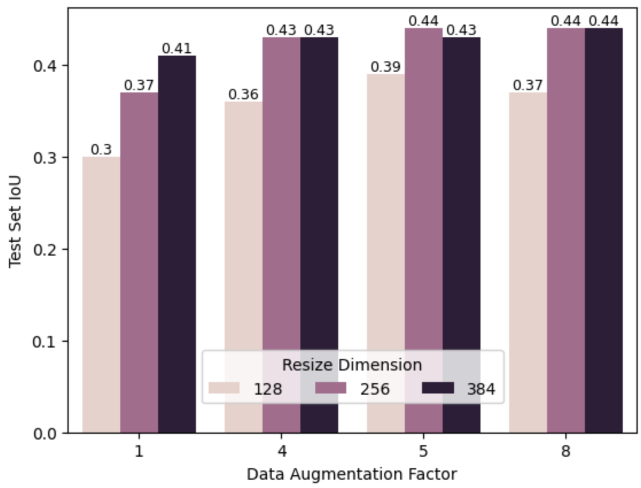

Figure 7: U-Net Bilinear Resizing Performance for Different Dimensions and Data Augmentation Factors

Under the bilinear resizing approach, the input images were resized to fixed dimensions of 128×128, 256×256, and 384×384 pixels. As shown in Figure 7, when increasing both the input size and the augmentation factor, the test performance consistently improved. The highest test IoU (0.44) was achieved at both 256×256 and 384×384 dimensions with an augmentation factor of 5 or 8.

However, this performance gain came at a computational cost. As seen in Figure 8, training time increased substantially with larger image sizes and heavier augmentation. For instance, the training time at 384×384 with an augmentation factor of 8 exceeded 90 minutes, compared to less than 20 minutes for 128×128. However, testing time was relatively stable between the difference of augmentation factor but still scaled with image size, with 384×384 requiring over 3.5 seconds, compared to ~1 second for 128×128.

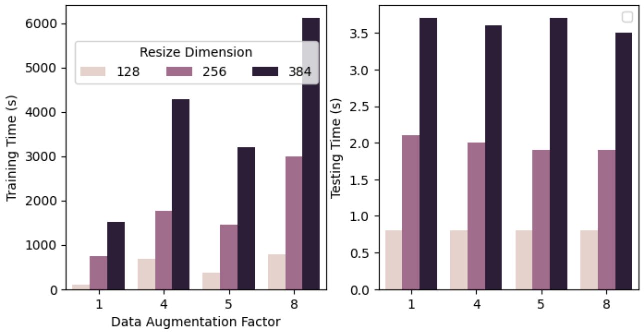

Figure 8: U-Net Bilinear Resizing Training and Testing Times

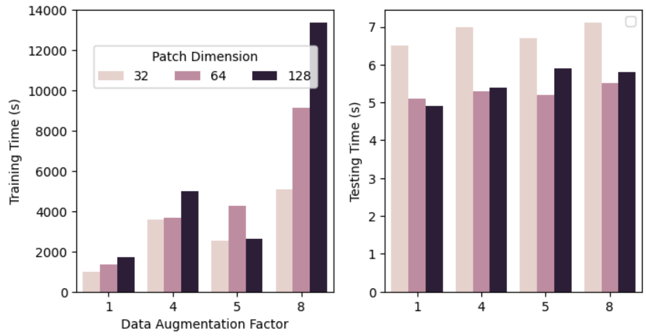

Figure 10: U-Net Patching Training and Testing Times

Patch Extraction Strategy:

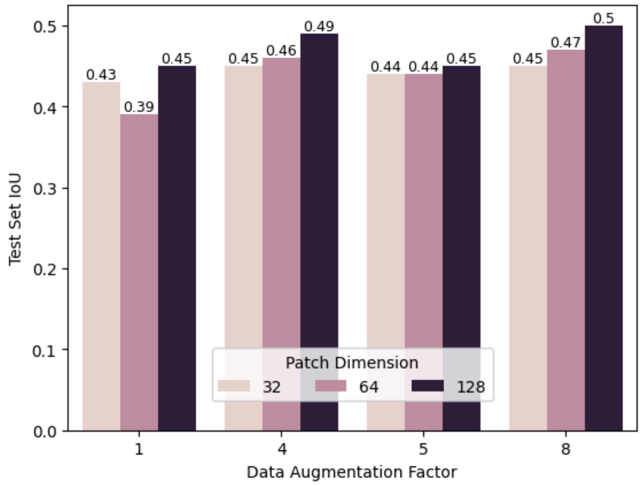

Figure 9: U-Net Patch Extraction Performance for Different Sized Patches and Data Augmentation Factors

For the patch-based approach, we extracted fixed-size patches of 32×32, 64×64, and 128×128 pixels. As shown in Figure 9, larger patches consistently outperformed smaller ones in terms of test IoU. The highest recorded performance was 0.50 at 128×128 with the maximum augmentation factor of 8. Patch extraction generally showed better IoU scores than bilinear resizing across all configurations.

However, the patching strategy performs significantly higher training time, especially at large patch sizes and augmentation levels. As shown in Figure 10, training time for 128×128 patches at an augmentation factor of 8 reached almost 4 hours which is nearly twice that of the largest resizing configuration. Nevertheless, the testing time was more stable across patch sizes, ranging from 5 to 7 seconds, slightly higher than that observed in the resized models.

C. Traditional Machine Learning Models

We conducted different experiments to evaluate how different hyperparameters and training set sizes affect the segmentation performance of traditional machine learning models. All experiments were conducted using 5-fold cross-validation with accuracy as the scoring metric, then we measured the final performance by using Mean Intersection over Union (IoU) on a fixed test set of 89 images after we found the best hyperparameter for each model.

For Random Forest, we tested a combination of n_estimators = [100, 200, 300] and max_depth = [None, 10, 20]. As shown in Figure 11, the best patch-wise classification accuracy was achieved when n_estimators is 300 and max_depth is None, achieving an accuracy of 0.92562 on the validation set. Which indicates that allowing trees to fully grow provided better results.

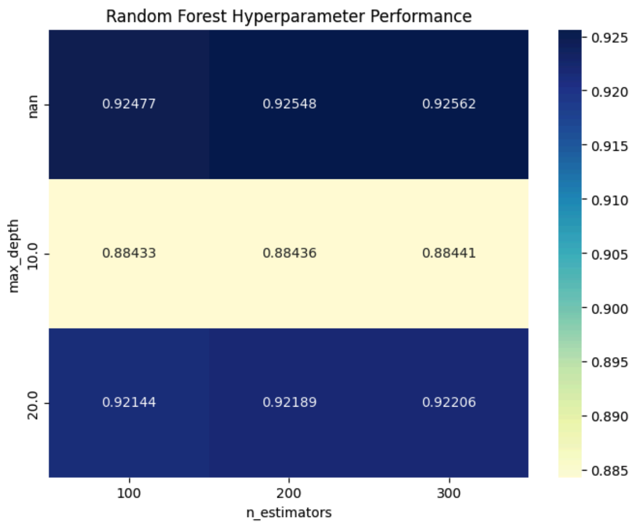

Figure 11: Validation accuracy of RandomForest with different max_depth and n_estimators

For k-Nearest Neighbors (k-NN), we evaluated n_neighbors = [3, 5, 7, 9]. Based on Figure 12, the model performed best when n_neighbors is 3, achieving an accuracy of approximately 0.915 on the validation set. Increasing the number of neighbors led to slight decrease in performance, with accuracy dropping to around 0.908 when n_neighbors is 9.

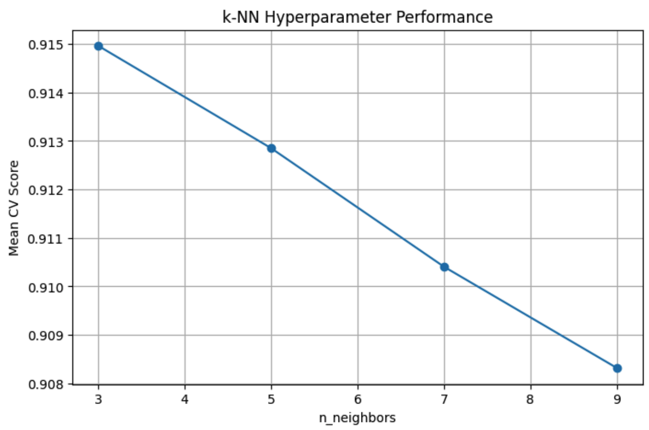

Figure 12: Validation accuracy of k-NN with different values of *n_neighbors*

For Support Vector Machine (SVM), we tested the regularisation parameter C of [0.5, 5, 10]. As shown in Figure 13, the model achieved the highest accuracy of 0.926 on the validation set when C is 10, and accuracy improved as the value of C increased.

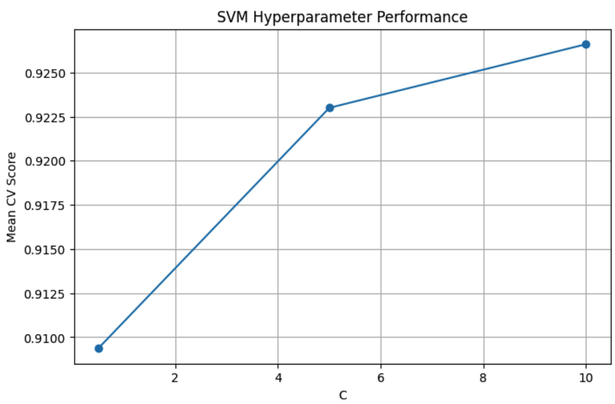

Figure 13: Validation accuracy of SVM with different values of C

In addition to hyperparameter tuning, we assessed how the size of the training set influences performance and computation time. According to the hardware limitation, we trained each model using a subset of 50 images and 250 images, while keeping the test set fixed. The results are summarised in Figure 14, which includes IoU scores as well as training and testing times.

Model	IoU	Training Time (s)	Testing Time (s)	Training Images	Testing Images
Random Forest	0.195551	6987.194296	2525.791971	250	89
k-NN	0.174465	6805.197348	2723.367922	250	89
SVM	0.215172	7840.327937	9750.637612	250	89
Random Forest	0.186166	1300.980706	2607.239957	50	89
k-NN	0.157471	1277.900964	2622.307085	50	89
SVM	0.211546	1293.883136	3596.219310	50	89

Figure 14: IoU, training and testing time when trained on 50 and 250 images

VI. Discussion

A. *Neural Methods*

Due to the limited size of the dataset (355 images in the training set), many of the techniques we experimented with hinged on attempting to mitigate this low-resource setting. Our ADA-Net adaptation used a pretrained ResNet-50 backbone in an attempt to transfer low-level feature identification from other classification tasks to this semantic segmentation task. Similarly by extracting fixed size patches we were able to increase the dataset by over 100-fold in the 32 x 32 patch case. By rotating and reflecting images we were also able to increase the dataset by 1, 4, 5 and 8 times and these effects stacked with patching. This huge increase in data is seen in the training time increases.

Data augmentation improved results across all model types, but with marginal returns as the dataset size increased, with the resizing strategy no longer benefitting from more than a 4 times increase in dataset size. This is likely due to the distortion of aspect ratios caused by resize, which if performed in multiple different directions (as the image is rotated), will make it difficult for the CNN to learn the features.

Similarly the ResNet-50 backbone clearly was trained on images with a consistent aspect ratio and by resizing images was not able to function as an effective decoder, achieving barely above random results. By contrast, by extracting patches we were able to preserve the original aspect ratio and performance was almost 5 times that of resizing.

Overall there were some inconsistencies in trends, such as a data augmentation of 4 being more effective than 5 for the patch extraction method. This may indicate that certain transformations adjust the appearance of trees in ways that are unnatural, but also is reflective of the fact that a single experiment was run for each configuration. The stochastic nature of the ADAM optimizer, shuffling of batches etc means that performance with the exact same dataset is likely to be variable. Batch sizes were also tuned mid experiment to improve runtime, as experiments took cumulatively multiple days, and this likely impacted the variation in results and training times. ADA-Net had the most noticeable variation in training time. Due to its higher number of parameters and no freezing of the backbone layers, it was able to overfit to the small validation set (8% of the original dataset size) and achieve train IoUs of up to 0.69. However, this overfitting did not consistently occur within the limited patience, hence the variation.

Maintaining a consistent patience of 8 epochs despite dramatically increasing the size of the training set may have led to an uneven increase in training for experiments with higher augmentation factors or use of patching. The differences may indicate that other strategies may be undertrained. Similarly, making the validation set 10% of the training set likely hurt models with little data augmentation or patching, as this is a validation set of just 35 images, making early stopping a less viable training method.

Overall it appears that despite the potential transfer learning has in other applications, our implementation did not perform as well as building and training a CNN specifically for our task. ResNet-50's image classification pretraining likely did not benefit the semantic segmentation task and it likely experienced catastrophic forgetting. The fixed input size of the ResNet-50 decoder also impacted our ability to utilise different strategies such as different patching sizes, which also made it a poor fit for our specific dataset.

B. *Traditional Machine Learning Models*

Across all three traditional machine learning models, Figure 15 shows that increasing the training size led to improvements in IoU. The k-NN model showed the largest relative improvement, while the SVM improved only slightly, suggesting that it may have already found the hyperplane with fewer training samples. However the performance gain came at a significant computational cost, especially for SVM. As shown in Figures 16 and Figure 17, the training time for SVM increased by more than sixfold and testing time more than tripled.

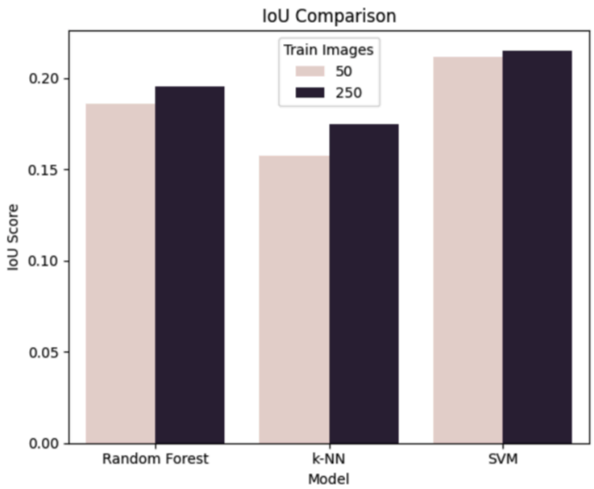

Figure 15: IoU performance comparison between 50 and 250 training images

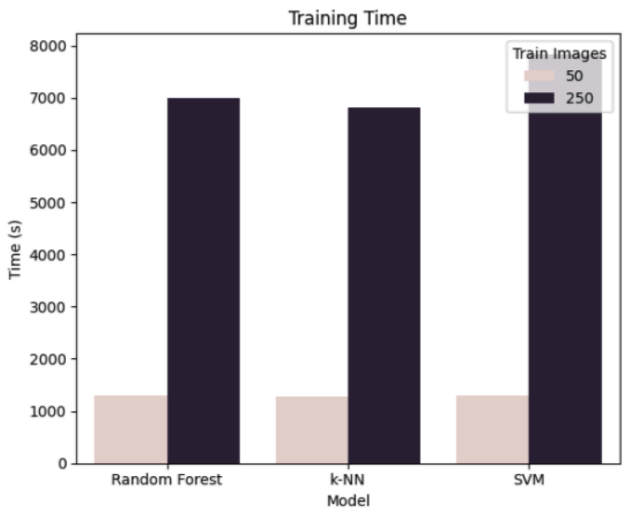

Figure 16: Training time comparison between 50 and 250 training images

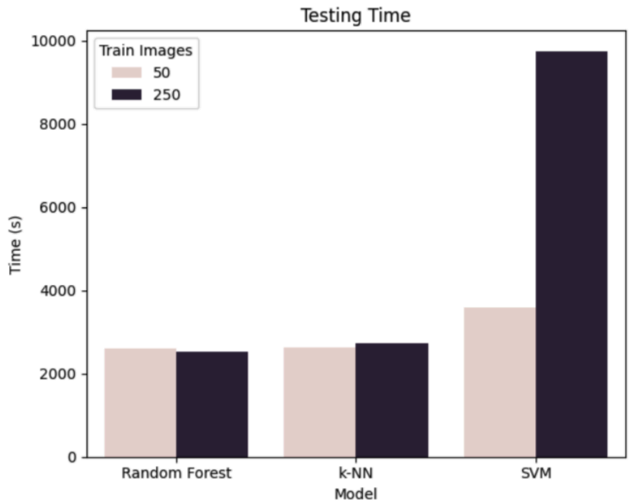

Figure 17: Testing time comparison between 50 and 250 training images

Ultimately the U-Net architecture, using patches of 128 x 128 of the images with a data augmentation factor of approximately 4 appears to have achieved the best tradeoff of IoU against computation time. Training time was approximately an hour less than traditional methods despite training on significantly more data, likely due to the fact that the CNNs could make use of CUDA and parallelise on the GPU far more effectively and because time consuming feature extraction was not required at test time. IoU performance was also more than double (0.21 vs 0.5) despite the much faster training and testing times.

VII. Conclusion

In this study, we compared traditional machine learning classifiers, ADA-Net and U-Net models for the task of dead tree segmentation. Traditional machine learning methods do not appear to be equipped for pixel level semantic segmentation, which requires a significant amount of feature creation and poorly utilises hardware. In contrast, deep learning architectures, particularly U-Net, showed a greater ability to capture spatial context and fine-grained details.

In future experiments, each neural configuration would benefit from multiple runs to better characterise the true mean and variation in performance, and give methods such as our ADA-Net adaptation an opportunity to escape initial local minima. Additional improvements could include using a decoder pretrained on semantic segmentation, some layer freezing and thawing and better finetuning of the learning rate could also help aspects of catastrophic forgetting and overfitting that appeared to occur in implementations using a backbone.

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